

[CH 12] The Internet

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In the 1990s, the internet experienced exponential growth

- 1990: 313,000 users
- 1995: almost 10 million
- End of 2000: already over 100 million

From the World Brain to the World Wide Web

There were 3 main desires that lead to the internet:

- A need for an efficient, fault-tolerant networking technology, suitable for military communications, that never broke down
- A wish to unite the world's computer networks into a single system
- To make readily available the world's store of knowledge

From the Encyclopedia to the Matrix

- The idea of making the world's knowledge available to everyone is the same that made Denis Diderot create the first great encyclopedia in the 18th century
 - The *Encyclopedie* was a central project of the enlightenment
 - The *Encyclopedia Britannica* (1768) was modelled after it
 - None of them could contain **ALL** the knowledge
 - They contained a significant portion of it, but they also gave people a sense of what and how much there was to know
- By the 1900s, knowledge was so ample that it was no longer possible to be knowledgeable in many topics. People had to specialize
- Before WW2, H.G.Wells envisioned the World Brain, a store of all of the world's information, sorted, summarised and analyzed by numerous workers.
 - He believed that it would not be centralized, but take the form of a network
 - He died one year after the end of the war, not managing to gather the needed support for his idea
- Vannevar Bush picked up the idea after WW2, when microfilm storage had advanced dramatically. His idea was the memex:
 - A whole library stored on microfilm inside a machine the size of a desk.
 - It allowed picking books by typing their ID on a keyboard and scrolling through them with levers
 - When he was making these predictions in 1945, the technology was still crude and too expensive
 - He was one of the few who saw in the computer more than an arithmetic machine
 - 1967, he thought it was still "in the future, but not so far"
- Ted Nelson and Douglas Engelbart claim to have been influenced by Dr. Vannevar Bush
- By then J.C.R. Licklider had already solved most of the intellectual problems involved in the project at the ARPA Information Processing Techniques Office
- The technology for the machine was there, but the network of computers would take a long time

The ARPANET

- 1963 the first ARPA-sponsored time-sharing systems became operational
- At the same time, Licklider set into action the Arpanet program
- The stated goal was economic: sharing the different ARPA centers' resources and more evenly distributing the computing load among them
- July 1964, when Licklider's 2-year term at the ARPA IPTO was over, he was followed by Ivan Sutherland (MIT graphics expert) and then Robert Taylor (MIT alumnus, later head of Xerox PARC)
- By 1966, the technology was ready for physical exploitation and Larry Roberts was appointed head of the project
- There were 3 problems
 - How to connect the computers to each other
 - How to use the expensive communication lines efficiently
 - How to link together all the different computer systems of various specs and from various manufacturers
- The solution to the first 2 already existed, unknown to Roberts: *store-and-forward packet switching* (Paul Baran, RAND Corporation, 1961), invented simultaneously by Donald Davies (National Physics Laboratory, England, 1965) as *packet switching*. This was similar to a telegraph technology, which used switching centers to avoid connecting every city to every other city
- To solve the third problem, he used the Interface Message Processor (IMP)
 - Instead of modifying existing software, a cheap minicomputer at every node would handle network traffic
- 1968 work began on the 2.5 million USD pilot scheme to connect 4 centers: University of California at Los Angeles, University of California at Santa Barbara, University of Utah and Stanford Research Institute
 - IMP software was contracted out to BBN

- 1970 it was fully operational and reliable
- By 1971 there were 23 hosts on the network
- To promote the idea, Roberts organized a demonstration at the International Conference on Computer Communications in Washington (1972)
- This left a great impression on attendees and was a turning point for Arpanet and networking in general

The Popularity of E-Mail

- Most users did not care much for video games, databases, remote access, etc. They were, however, very interested in communicating through electronic mail
- By 1975 there were over 1,000 email users
- The demand for email was also a major driving force for non-ARPA networks
- Usenet was a network for colleges that were excluded from Arpanet.
 - By 1991 there were 35,000 usenet nodes and millions of subscribers to news groups
- Time sharing firms began to restructure themselves as network providers: BBN Telcomp became Telnet, Tymshare created Tymnet as a subsidiary, Western Union MCI, NSF, NASA started providing email and networks, universities created networks like Merit, EduNet, Bitnet
- To prevent proprietary networks from inhibiting communications between computers, IPTO set out to create network protocols
- ARPA developed TCP/IP, which is still the standard today
- 1980 there were less than 200 hosts on the Internet
- 1984 there were only 1,000
- Late 1980s most professional computer users had internet access

The World Wide Web

- With more and more people gaining access to the internet, the amount of information and documents skyrocketed
- Sorting through this and finding the right items became a major research issue
 - McGill University created "archie" in 1990. It combed through the internet and created a directory of all the downloadable files on the internet
 - Thinking Machines Corporation of Waltham, Massachusetts created WAIS (Wide Area Information System), which would search for documents that matched user-specified keywords
 - University of Minnesota created "gopher" in 1991. It was a "catalog of catalogs". It let users examine the contents of gopher databases maintained by hundreds of institutions. (this was the most successful one)
- Individual documents were treated like books. If someone wanted to view any other different document, they would have to search again.
- Hypertext was meant to improve this, by allowing jumps between documents through links stored within the documents. This made it no longer necessary to have centralized directories
- The World Wide Web was invented by Tim Berners-Lee
- While working at CERN, he developed a Hypertext system called Enquire, but it never got any traction
- After a short break and returning to CERN, he was tasked with helping connect all of the institution's computers
 - For this, he dusted off project Enquire and his interest in hypertext
 - 1989, Berners-Lee and his collaborator Robert Caillau made a formal project proposal to CERN for the World Wide Web
- WWW had 2 sides:
 - Server side: deliver hypertext documents to clients
 - Client side: display the pages to the user
- 1991 there was little interest at conferences
- By 1993, it was everywhere. Universities, companies and individuals were developing browser software and server programs
- Berners-Lee pushed for CERN to place web technology in the public domain so that it could be freely used by anyone
- 1994 he moved to MIT where he headed the W3C (World Wide Web Consortium), a nonprofit encouraging the creation of standards

The Web and its Consequences

- 1993 the Mosaic browser, developed by 22-year-old undergraduate computer science student Marc Andreessen
- Unlike most other browsers, it had a finished feel, almost like store-bought shrink-wrapped software

Browser Wars

- Andreessen and Jim Clark (co-founder of Silicon Graphics) incorporated Mosaic Communications Corporation (1994), renamed to Netscape Communications months later
- December 1994 Netscape was made available and was downloaded by the millions
- By mid 1995, the web accounted for a quarter of internet traffic

- 1995 Netscape had 70% of the market, but Microsoft did not plan to concede
- For 2 years, Netscape and Internet Explorer battled
- As Internet Explorer came with Windows 98, it became the most used browser. It was also made available for free for the Macintosh
- When Microsoft was giving the browser away for free, Netscape was having trouble selling a browser

The Internet Landrush

- 1990 the internet was largely owned by US Government agencies
- The National Science Foundation created NSFnet and it became the backbone of the internet
- By 1987 30,000 computers had access to the Internet
- Public and private networks attached themselves (Usenet, FidoNet, Bitnet (by IBM))
- 1995 the NSFnet backbone was shut down, ending US government ownership of the Internet's infrastructure
- Mid 1980s DNS was adopted, designed by Paul Mockapetris
- Electronic Commerce on the internet was something somewhat unexpected
- Early successes were sometimes accidental (like Yahoo!)
- Monetisation was done through advertising, subscriptions, sponsorships, etc
- Yahoo! Had employees manually sift through new websites and index them
- While working on the Stanford Digital Library Project, Larry Page and Sergey Brin used a web crawler to create "PageRank".
 - Pagerank is an algorithm based on back-links (websites that link to a particular site)
- September 1997 they launched Google (renaming it from Backrub)
- 1998 they launched Google Inc. in a Menlo Park garage
- Early 1999 they moved to Palo Alto
- 2000 Google had a loyal following
- By the early 2000s it was the leading search engine
- They were expanding, hiring more staff, taking loans, refining technology, extending server infrastructure
- 2001 they hired professional CEO Eric Schmidt
- Google perfected sponsored search
- 2004 the Google public offering valued the company at over \$26 Billion
- 1995 Jeff Bezos launched Amazon.com, which set many of the web commerce's early practices
- Initially, it was an online bookseller, but it became the world's top online retailer
- eBay was developed by Pierre Omidyar in late 1995
- By 1996 it was popular enough that he could start charging commissions

Going Mobile

- With the advent of PCs, computers have become more and more mobile
- 1972, Alan Kay conceptualized the Dynabook at Xerox PARC
- 1982 GRiD launched the GRiD Compass 1101, the first laptop computer
- As microprocessors, microchips and screen technology progressed, laptops became more powerful and more portable
- They were still too large to be a personal accessory
- This place was taken by the smartphone
- Before 2007: Symbian, BlackBerry OS, Palm OS and Windows Mobile
- Since then: Apple's iOS and Google's Android
- The predecessor of the Smartphone is the PDA (Personal Digital Assistant)
- 1986 Psion introduced the Psion Organizer II, an OS based PDA
- The EPOC OS was developed into Symbian after Psion formed a consortium with Nokia
- 2007, 2/3 of smartphones were based on Symbian
- Competition and their features started eroding the Symbian market lead
- 1999 RIM launched the "Blackberry" PDA
- Mid 1980s, Apple Computer was struggling and Steve Jobs was voted off the management team and resigned
- He started NeXT, which acquired the computer graphics division of Lucasfilms and spun it off as Pixar. This made Jobs a billionaire
- 1997 he was invited to lead Apple Computer
- He envisioned Apple selling consumer electronics
- He discontinued the unsuccessful Apple Newton PDA
- Apple launched the iPod, whose success was compounded by the simultaneous launch of the iTunes store
- 2007 the iPhone was launched

- 2005 Google acquired Android Inc.
- They made sure to keep it open-source and royalty free
- A vast ecosystem of apps emerged

Web 2.0

- In the second half of the 1990s, those creating web content were few
- In the early 2000s, user content was encouraged